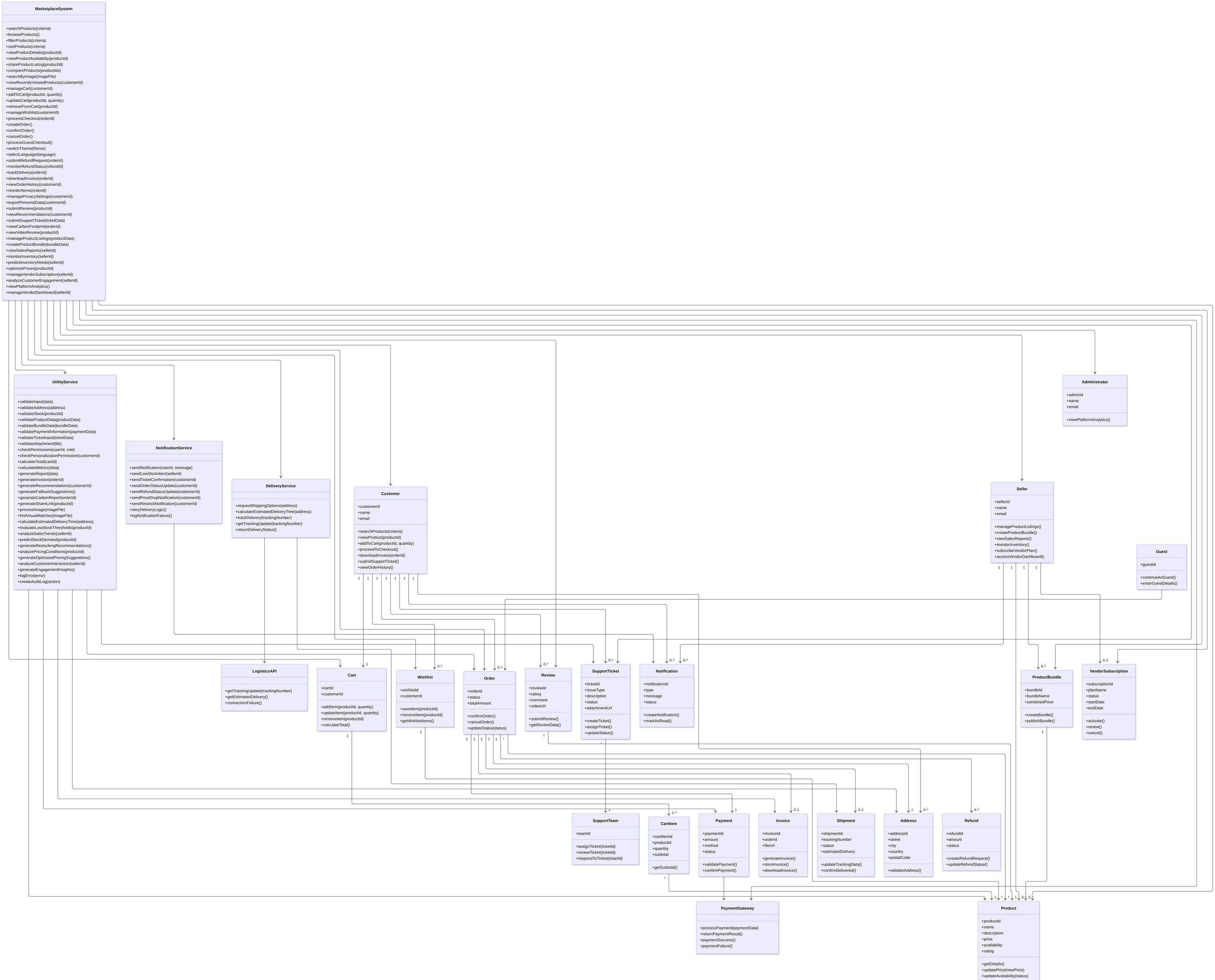
MarketplaceSystem
+searchProducts(criteria)
+browseProducts()
+filterProducts(criteria)
+sortProducts(criteria)
+viewProductDetails(productId)
+viewProductAvailability(productId)
+shareProductListing(productId)
+compareProducts(productIds)
+searchByImage(imageFile)
+viewRecentlyViewedProducts(customerId)
+manageCart(customerId)
+addToCart(productId, quantity)
+updateCart(productId, quantity)
+removeFromCart(productId)
+manageWishlist(customerId)
+processCheckout(orderId)
+createOrder()
+confirmOrder()
+cancelOrder()
+processGuestCheckout()
+switchTheme(theme)
+selectLanguage(language)
+submitRefundRequest(orderId)
+monitorRefundStatus(refundId)
+trackDelivery(orderId)
+downloadInvoice(orderId)
+viewOrderHistory(customerId)
+reorderItems(orderId)
+managePrivacySettings(customerId)
+exportPersonalData(customerId)
+submitReview(productId)
+viewRecommendations(customerId)
+submitSupportTicket(ticketData)
+viewCarbonFootprint(orderId)
+viewVideoReview(productId)
+manageProductListings(productData)
+createProductBundle(bundleData)
+viewSalesReports(sellerId)
+monitorInventory(sellerId)
+predictInventoryNeeds(sellerId)
+optimizePrices(productId)
+manageVendorSubscription(sellerId)
+analyzeCustomerEngagement(sellerId)
+viewPlatformAnalytics()
+manageVendorDashboard(sellerId)
UtilityService
+validateInput(data)
+validateAddress(address)
+validateStock(productId)
+validateProductData(productData)
+validateBundleData(bundleData)
+validatePaymentInformation(paymentData)
+validateTicketInput(ticketData)
+validateAttachment(file)
+checkPermissions(userId, role)
+checkPersonalizationPermission(customerId)
+calculateTotal(cartId)
+calculateMetrics(data)
+generateReport(data)
+generateInvoice(orderId)
+generateRecommendations(customerId)
+generateFallbackSuggestions()
+generateCarbonReport(orderId)
+generateShareLink(productId)
+processImage(imageFile)
+findVisualMatches(imageFile)
+calculateEstimatedDeliveryTime(address)
+evaluateLowStockThresholds(productId)
+analyzeSalesTrends(sellerId)
+predictStockDemand(productId)
+generateRestockingRecommendations()
+analyzePricingConditions(productId)
+generateOptimizedPricingSuggestions()
+analyzeCustomerInteraction(sellerId)
+generateEngagementInsights()
+logError(error)
+createAuditLog(action)
NotificationService
+sendNotification(userId, message)
+sendLowStockAlert(sellerId)
+sendTicketConfirmation(customerId)
+sendOrderStatusUpdate(customerId)
+sendRefundStatusUpdate(customerId)
+sendPriceDropNotification(customerId)
+sendRestockNotification(customerId)
+retryDeliveryLogic()
+logNotificationFailure()
DeliveryService
+requestShippingOptions(address)
+calculateEstimatedDeliveryTime(address)
+trackDelivery(trackingNumber)
+getTrackingUpdate(trackingNumber)
+returnDeliveryStatus()
Customer
+customerId
+name
+email
+searchProducts(criteria)
+viewProduct(productId)
+addToCart(productId, quantity)
+proceedToCheckout()
+downloadInvoice(orderId)
+submitSupportTicket()
+viewOrderHistory()
Administrator
+adminId
+name
+email
+viewPlatformAnalytics()
Seller
+sellerId
+name
+email
+manageProductListings()
+createProductBundle()
+viewSalesReports()
+monitorInventory()
+subscribeVendorPlan()
+accessVendorDashboard()
Guest
+guestId
+continueAsGuest()
+enterGuestDetails()
LogisticsAPI
+getTrackingUpdate(trackingNumber)
+getEstimatedDelivery()
+connectionFailure()
Cart
+cartId
+customerId
+addItem(productId, quantity)
+updateItem(productId, quantity)
+removeItem(productId)
+calculateTotal()
Wishlist
+wishlistId
+customerId
+saveItem(productId)
+removeItem(productId)
+getWishlistItems()
Order
+orderId
+status
+totalAmount
+confirmOrder()
+cancelOrder()
+updateStatus(status)
Review
+reviewId
+rating
+comment
+videoUrl
+submitReview()
+getReviewData()
SupportTicket
+ticketId
+issueType
+description
+status
+attachmentUrl
+createTicket()
+assignTicket()
+updateStatus()
Notification
+notificationId
+type
+message
+status
+createNotification()
+markAsRead()
ProductBundle
+bundleId
+bundleName
+combinedPrice
+createBundle()
+publishBundle()
VendorSubscription
+subscriptionId
+planName
+status
+startDate
+endDate
+activate()
+renew()
+cancel()
SupportTeam
+teamId
+assignTicket(ticketId)
+reviewTicket(ticketId)
+respondToTicket(ticketId)
CartItem
+cartItemId
+productId
+quantity
+subtotal
+getSubtotal()
Payment
+paymentId
+amount
+method
+status
+validatePayment()
+confirmPayment()
Invoice
+invoiceId
+orderId
+fileUrl
+generateInvoice()
+storeInvoice()
+downloadInvoice()
Shipment
+shipmentId
+trackingNumber
+status
+estimatedDelivery
+updateTrackingData()
+confirmDelivered()
Address
+addressId
+street
+city
+country
+postalCode
+validateAddress()
Refund
+refundId
+amount
+status
+createRefundRequest()
+updateRefundStatus()
PaymentGateway
+processPayment(paymentData)
+returnPaymentResult()
+paymentSuccess()
+paymentFailure()
Product
+productId
+name
+description
+price
+availability
+rating
+getDetails()
+updatePrice(newPrice)
+updateAvailability(status)